Return your solutions on Moodle before **Sunday 15.3. at 23:59**. Your solutions must include intermediate steps and explain your reasoning.

Exercise 2.1. Compute (i) the absolute error, (ii) the relative error, and (iii) and the precision of approximation when $x = 0.5772156649$ is approximated with

(a) $\tilde{x} = 0.7$

(b) $\tilde{x} = 0.55$

(c) $\tilde{x} = 0.577$

(d) $\tilde{x} = 0.57701$

(e) $\tilde{x} = 0.57721$

(f) $\tilde{x} = \pi$

Solution. Recall that (i) *absolute error* $:= |x - \tilde{x}|$, (ii) *relative error* $:= \frac{|x-\tilde{x}|}{|x|}$, (iii) and (precision of an approximation) $\tilde{x}$ is said to approximate x to d decimal places if $|x - \tilde{x}| < \frac{1}{2}10^{-d}$.

Note that $|x - \tilde{x}| < \frac{1}{2}10^{-d} \iff d < -\log_{10}(2|x - \tilde{x}|)$, where $d \in \mathbb{N}$: if $\tilde{x}$ approximates x to d decimal places, then $\tilde{x}$ also approximates x to $d-1$ decimal places, $d-2$ decimal places, and so on. We define $d_{\max} := \max\{d \in \mathbb{N} \mid d < -\log_{10}(2|x - \tilde{x}|)\}$ as the representative of the set of solutions (that if $\tilde{x}$ approximates x to $d_{\max} = 5$ decimal places, then $\tilde{x}$ also approximates x to $d_{\max} - 1 = 4$ decimal places, $d_{\max} - 2 = 3$ decimal places, etc).

Firstly, a result useful for all sub-tasks: $|x| = |0.5772156649| = 0.5772156649$.

(a) $\tilde{x} = 0.7$

i. $|x - \tilde{x}| = |0.5772156649 - 0.7| = |-0.1227843351| = 0.1227843351$

ii. $\frac{|x-\tilde{x}|}{|x|} = \frac{0.1227843351}{0.5772156649} \approx 0.2127183 \approx 21.27\%$

iii. $d < -\log_{10}(2|x - \tilde{x}|) = -\log_{10}(2 \cdot 0.1227843351) \approx 0.609827$, since $d \in \mathbb{N}$ we get $d = 0$, so $d_{\max} = 0$

(b) $\tilde{x} = 0.55$

i. $|x - \tilde{x}| = |0.5772156649 - 0.55| = |0.0272156649| = 0.0272156649$

ii. $\frac{|x-\tilde{x}|}{|x|} = \frac{0.0272156649}{0.5772156649} \approx 0.0471499 \approx 4.71\%$

iii. $d < -\log_{10}(2|x - \tilde{x}|) = -\log_{10}(2 \cdot 0.0272156649) \approx 1.2641511$, since $d \in \mathbb{N}$ we get $d \leq 1$, so $d_{\max} = 1$

(c) $\tilde{x} = 0.577$

i. $|x - \tilde{x}| = |0.5772156649 - 0.577| = |0.0002156649| = 0.0002156649$

ii. $\frac{|x-\tilde{x}|}{|x|} = \frac{0.0002156649}{0.5772156649} \approx 0.0003736 \approx 0.04\%$

iii. $d < -\log_{10}(2|x - \tilde{x}|) = -\log_{10}(2 \cdot 0.0002156649) \approx 3.3651905$, since $d \in \mathbb{N}$ we get $d \leq 3$, so $d_{\max} = 3$

(d) $\tilde{x} = 0.57701$

i. $|x - \tilde{x}| = |0.5772156649 - 0.57701| = |0.0002056649| = 0.0002056649$

ii. $\frac{|x-\tilde{x}|}{|x|} = \frac{0.0002056649}{0.5772156649} \approx 0.0003563 \approx 0.04\%$

iii. $d < -\log_{10}(2|x-\tilde{x}|) = -\log_{10}(2 \cdot 0.0002056649) \approx 3.3858098$, since $d \in \mathbb{N}$ we get $d \leq 3$, so $d_{\max} = 3$

(e) $\tilde{x} = 0.57721$

i. $|x-\tilde{x}| = |0.5772156649 - 0.57721| = |0.0000056649| = 0.0000056649$

ii. $\frac{|x-\tilde{x}|}{|x|} = \frac{0.0000056649}{0.5772156649} \approx 0.0000098 \approx 0.001\%$

iii. $d < -\log_{10}(2|x-\tilde{x}|) = -\log_{10}(2 \cdot 0.0000056649) \approx 4.9457778$, since $d \in \mathbb{N}$ we get $d \leq 4$, so $d_{\max} = 4$

(f) $\tilde{x} = \pi$

i. $|x-\tilde{x}| = |0.5772156649 - \pi| \approx |-2.5643769886| = 2.5643769886$

ii. $\frac{|x-\tilde{x}|}{|x|} \approx \frac{2.5643769886}{0.5772156649} \approx 4.4426670 \approx 444.27\%$

iii. $d < -\log_{10}(2|x-\tilde{x}|) \approx -\log_{10}(2 \cdot 2.5643769886) \approx -0.7100119$, since $d \in \mathbb{N}$ the inequality has no solution and we say that $\tilde{x}$ does not approximate x to any decimal place.

□

Exercise 2.2. For the following pairs of functions f and g, determine if $f(x) = O(g(x))$ or $g(x) = O(f(x))$ as $x \to 0$:

(a) $f(x) = 1$ and $g(x) = x^2$.

(b) $f(x) = x^2$ and $g(x) = x + x^2$.

(c) $f(x) = x^2 + x^4$ and $g(x) = x^2$.

(d) $f(x) = e^x - 1$ and $g(x) = x^2$.

Solution. Your solutions for this problem are not expected to be nearly as detailed as the model solutions. Moreover, $f(x) = O(g(x))$ in (b) and $g(x) = O(f(x))$ in (d) are accepted as correct answers if you implicitly assumed that $x > 0$. Let's recall the definition of Big-O notation for functions. Let $x_0 \in (a,b)$ and let $f, g : (a,b) \to \mathbb{R}$ be functions. We say $f(x) = O(g(x))$ as $x \to x_0$ if there exist $C \geq 0$ and $\delta > 0$ such that $|f(x)| \leq Cg(x)$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

Additionally, recall the triangle inequality: $|x+y| \leq |x| + |y|$, for all $x, y \in \mathbb{R}$. And recall that $|x|^n \leq |x|^m$ if $|x| \leq 1$ and $n \geq m \geq 0$.

Since $x \to 0$, therefore $x_0 = 0$: thus the interval of interest is $x \in (0-\delta, 0+\delta) = (-\delta, \delta)$, for some $\delta > 0$.

(a) $f(x) = 1$ and $g(x) = x^2$.

- If $f(x) = O(g(x))$, then it must hold that $|f(x)| \leq Cg(x)$ for some $C \geq 0$, i.e. $|1| \leq Cx^2$. It's easy to show that the previous inequality has no solution: i.e. there is no C s.t. $Cx^2 \geq 1$ for all $x \in (-\delta, \delta)$. For example take $x = 0$, then $C \cdot 0^2 = 0 \geq 1$, which is false. Therefore $f(x) \neq O(g(x))$.

- Choose $\delta = 1$, therefore $|x| < 1$ for all $x \in (-\delta, \delta) = (-1, 1)$. Now, observe that

$$|g(x)| = |x^2| = |x|^2 \leq |x|^0 = 1 = f(x),$$

therefore, with $C = 1$, we have $g(x) = O(f(x))$.

(b) $f(x) = x^2$ and $g(x) = x + x^2$.

- If $f(x) = O(g(x))$, then it must hold that $|f(x)| \leq Cg(x)$ for some $C \geq 0$, i.e. $|x^2| \leq C(x+x^2)$. Restrict $\delta \leq 1$ (we are interested to find at least one interval $(x_0 - \delta, x_0 + \delta)$ containing x_0, no matter how small it is). Take $a \in (0, \delta)$. Since the previous inequality must hold for all $x \in (-\delta, \delta)$, it also must hold for $x = -a$, which has the following implication for C:

$$\begin{aligned} |(-a)^2| &\leq C((-a) + (-a)^2) \\ |a^2| &\leq C(-a + a^2) \\ a^2 &\leq C(-a + a^2) \end{aligned}$$

Observe that $-a + a^2 < 0 \iff a > a^2 \iff 1 > a$ is verified since by definition $0 < a < \delta \leq 1$, therefore when diving by $(-a + a^2)$ the direction of the inequality changes

$$\frac{a^2}{a^2 - a} \geq C$$

Now observe that, since $a^2 > 0$ and $-a + a^2 < 0$, we have $\frac{a^2}{a^2 - a} < 0$, which implies that $C < 0$. Hence $|f(x)| \leq Cg(x)$ has no solution for $C \geq 0$. Therefore $f(x) \neq O(g(x))$.

- If $g(x) = O(f(x))$, then it must hold that $|g(x)| \leq Cf(x)$ for some $C \geq 0$, i.e. $|x + x^2| \leq Cx^2$. Suppose that the previous inequality has solution for some $C = k$ for all $x \in (-\delta, \delta)$, then

$$|x + x^2| \leq kx^2$$

Let's first study the case $x \in (0, \delta)$, then $x + x^2 > 0$ since $x > 0$, therefore

$$\begin{aligned} x + x^2 &\leq kx^2 \\ kx^2 - x^2 - x &\geq 0 \\ (k-1)x^2 - x &\geq 0 \\ x((k-1)x - 1) &\geq 0 \end{aligned}$$

Since $x > 0$ we can divide by x both sides and the direction of the inequality does not change

$$(k-1)x - 1 \geq 0$$

If $k > 1$:

$$x \geq \frac{1}{k-1}$$

then take $x \in (0, \frac{1}{k-1})$, and observe that the inequality $|x + x^2| \leq kx^2$ does not hold anymore (if $\frac{1}{k-1} \geq \delta$ then it never held).

Else, if $k = 1$:

$$\begin{aligned}(1-1)x - 1 &\geq 0\\ 0x - 1 &\geq 0\\ -1 &\geq 0\end{aligned}$$

which is false.
Else, if $k < 1$:

$$x \leq \frac{1}{k-1}$$

But observe that since $k-1 < 0$ then $\frac{1}{k-1} < 0$, therefore $x < 0$, which contradicts the assumption that $x \in (0, \delta)$. Therefore it is false, in all cases.
Since $C = k$ was arbitrary, there's no C s.t. $|g(x)| \leq Cf(x)$. Therefore $g(x) \neq O(f(x))$.

(c) $f(x) = x^2 + x^4$ and $g(x) = x^2$. Choose $\delta = 1$, therefore $|x| < 1$ for all $x \in (-\delta, \delta) = (-1, 1)$.

- Observe that

$$|f(x)| = |x^2 + x^4| \leq |x^2| + |x^4| = |x|^2 + |x|^4 \leq |x|^2 + |x|^2 = 2|x|^2 = 2x^2 = 2g(x)$$

  therefore, with $C = 2$, we have $f(x) = O(g(x))$.
- But also note that

$$|g(x)| = |x^2| = x^2 \leq x^2 + x^4 = f(x)$$

  where we use the fact that $x^4 \geq 0$. Therefore, with $C = 1$, we have $g(x) = O(f(x))$.

(d) $f(x) = e^x - 1$ and $g(x) = x^2$.

- If $f(x) = O(g(x))$, then it must hold that $|f(x)| \leq Cg(x)$ for some $C \geq 0$, i.e. $|e^x - 1| \leq Cx^2$. Suppose there exist a solution for some $C = k$. Since it must hold for all $x \in (-\delta, \delta)$, let's first study the case $x \in (0, \delta)$. Recall that e^x is monotonic increasing. Since $e^0 = 1$, it follows that $e^x > 1$ for all $x > 0$, therefore $e^x - 1 > 0$ for all $x \in (0, \delta)$:

$$\begin{aligned}|e^x - 1| &\leq kx^2\\ e^x - 1 &\leq kx^2\\ kx^2 - e^x + 1 &\geq 0\end{aligned}$$

  Define $d(x) = kx^2 - e^x + 1$, i.e. the distance between kx^2 and $e^x - 1$ as a function of x. As the previous inequality tells, it must always be greater or equal than zero, i.e. $d(x) \geq 0$, otherwise $C = k$ wouldn't be a valid solution. Compute the first derivative $d'(x) = k \cdot 2x - e^x$. Observe that

$$d'(0) = k \cdot 0 - e^0 = -1$$

  which means that there exist a small enough ε s.t. $d(0) > d(0 + \varepsilon)$ (since the derivative is negative), with $0 + \varepsilon \in (0, \delta)$ for all δ. Now observe that

$$d(0) = k \cdot 0 - e^0 + 1 = -1 + 1 = 0$$

  Therefore we get that $0 > d(0 + \varepsilon)$, which contradicts that $d(x) \geq 0$ for all $x \in (0, \delta)$. Since $C = k$ was arbitrary, there's no C s.t. $e^x - 1 \leq Cx^2$ for all $x \in (0, \delta)$, i.e. there's no C s.t. $|e^x - 1| \leq Cx^2$ for all $x \in (-\delta, \delta)$. Therefore $f(x) \neq O(g(x)$.

- If $g(x) = O(f(x))$, then it must hold that $|g(x)| \leq Cf(x)$ for some $C \geq 0$, i.e. $|x^2| \leq C(e^x - 1)$. Since it must hold for all $x \in (-\delta, \delta)$, let's first study the case $x \in (-\delta, 0)$. Recall that e^x is monotonic increasing. Since $e^0 = 1$, it follows that $e^x < 1$ for all $x < 0$, therefore $e^x - 1 < 0$ for all $x \in (-\delta, 0)$. Since $C \geq 0$, we have that $C(e^x - 1) \leq 0$, but $|x^2| > 0$ for all $x \in (-\delta, 0)$. Therefore

$$0 < |x^2| \leq C(e^x - 1) \leq 0 \iff 0 < 0$$

which is false. Hence $|g(x)| \leq Cf(x)$ has no solution and thus $g(x) \neq O(f(x))$.

□

Exercise 2.3. The *central difference method* approximates the derivative at x_0 as follows:

$$f'(x_0) \approx \mathrm{cd}(h) = \frac{f(x_0 + h) - f(x_0 - h)}{2h}.$$

Determine the order of the absolute error $|f'(x_0) - \mathrm{cd}(h)|$ of the central difference method as $h \to 0$ for the function $f(x) = x^3$ and point $x_0 = 2$.

Solution. Since $f(x) = x^3$, the exact derivative is $f'(x) = 3x^2$. Therefore the absolute error at $x_0 = 2$ is

$$\begin{aligned}
\left|f'(x_0) - \mathrm{cd}(h)\right| &= \left|f'(2) - \frac{f(2+h) - f(2-h)}{2h}\right| \\
&= \left|3 \cdot 2^2 - \frac{(2+h)^3 - (2-h)^3}{2h}\right| \\
&= \left|12 - \frac{(8 + 12h + 6h^2 + h^3) - (8 - 12h + 6h^2 - h^3)}{2h}\right| \\
&= \left|12 - \frac{24h + 2h^3}{2h}\right| \\
&= \left|12 - \frac{2h(12 + h^2)}{2h}\right| \\
&= \left|12 - (12 + h^2)\right| \\
&= |h^2| \\
&= h^2
\end{aligned}$$

Which shows that $|f'(x_0) - \mathrm{cd}(h)| = O(h^2)$ at $x_0 = 2$. Note that to reach the result $O(h^2)$ (i.e. the order of the absolute error for *central difference method* to approximate $f'(x)$ for $f(x) = x^3$ at $x_0 = 2$) we didn't use the fact that $h \to 0$: because the absolute error simplifies to be exactly equal to h^2, it will be of order $O(h^2)$ as $h \to h_0$ for any h_0, not just $h_0 = 0$, since $|h^2| \leq Ch^2$ for all $C \geq 1$ and $h \in \mathbb{R}$. □

Exercise 2.4. Prove the following claims:

(a) If $a_n = O(b_n)$ and $b_n = O(c_n)$ as $n \to \infty$, then $a_n = O(c_n)$ as $n \to \infty$.

(b) $e^{-n} = O(n^{-p})$ as $n \to \infty$ for any $p > 0$.

Solution. Let's recall the definition of Big-O notation for sequences. Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences. We write $a_n = O(b_n)$ as $n \to \infty$ if there is $C \geq 0$ such that $|a_n| \leq Cb_n$ for all $n \geq 1$.

(a) *Claim.* If $a_n = O(b_n)$ and $b_n = O(c_n)$ as $n \to \infty$, then $a_n = O(c_n)$ as $n \to \infty$.

Proof. By definition, there are $C_1, C_2 \geq 0$ such that

$$|a_n| \leq C_1 b_n \quad \text{for all} \quad n \geq 1$$

and

$$|b_n| \leq C_2 c_n \quad \text{for all} \quad n \geq 1$$

These two inequalities are valid simultaneously. Therefore, for all $n \geq 1$,

$$|a_n| \leq C_1 b_n \leq C_1 |b_n| \leq C_1 C_2 c_n,$$

Set $C = C_1 \cdot C_2$ and we get that $|a_n| \leq Cc_n$ for all $n \geq 1$, i.e. $a_n = O(c_n)$ as $n \to \infty$.

(b) *Claim.* $e^{-n} = O(n^{-p})$ as $n \to \infty$ for any $p > 0$.

Proof. To prove the claim we need to find $C \geq 0$ such that $|e^{-n}| \leq Cn^{-p}$ for all $n \geq 1$.

$$|e^{-n}| \leq Cn^{-p}$$

since $e^x > 0$ for all $x \in \mathbb{R}$

$$e^{-n} \leq Cn^{-p}$$

since $n \geq 1$, also $n > 0$, therefore $n^x > 0$ for all $x \in \mathbb{R}$ (i.e we can multiply both sides by n^p without changing the direction of the inequality sign)

$$n^p e^{-n} \leq C$$
$$C \geq \frac{n^p}{e^n}$$

The previous claim must hold for all $n \geq 1$. Therefore we could set $C = \max\{\frac{n^p}{e^n} \mid n \geq 1\}$ and the claim would always hold. By computing the derivative of $\frac{n^p}{e^n}$ in the direction of n we have

$$\frac{d}{dn}\left(\frac{n^p}{e^n}\right) = (pn^{-1} - 1)n^p e^{-n}$$

Now let's study the sign of the derivative

$$(pn^{-1} - 1)n^p e^{-n} \geq 0$$
$$pn^{-1} - 1 \geq 0$$
$$pn^{-1} \geq 1$$
$$p \geq n$$
$$n \leq p$$

where we used the fact that $n^p e^{-n} > 0$ and $n > 0$. The latter result means that the derivative

$$\begin{aligned}
\frac{d}{dn}\left(\frac{n^p}{e^n}\right) &> 0 \quad \text{when} \quad n < p \\
\frac{d}{dn}\left(\frac{n^p}{e^n}\right) &= 0 \quad \text{when} \quad n = p \\
\frac{d}{dn}\left(\frac{n^p}{e^n}\right) &< 0 \quad \text{when} \quad n > p
\end{aligned}$$

As a basic calculus result, $n_{\max} = p$ is a maxima point, i.e. $\max\{\frac{n^p}{e^n} \mid n \geq 1\} = \frac{p^p}{e^p}$, when $p \geq 1$ (the latter constraint is a consequence that $n \geq 1$). But, when $p < 1$, then, since the first derivative would always be negative (within the domain $n \geq 1$), the maxima point would be at the domain boundary, i.e. $n_{\max} = 1$, with $\max\{\frac{n^p}{e^n} \mid n \geq 1\} = \frac{1^p}{e^1} = \frac{1}{e}$. Therefore, choose $C_{\max}$ as $\frac{n^p}{e^n}$ when $p \geq 1$, or $C_{\max}$ as $\frac{1}{e}$ when $p < 1$. Then $|e^{-n}| \leq C_{\max} n^{-p}$ for all $n \geq 1$, i.e. $e^{-n} = O(n^{-p})$ as $n \to \infty$ for any $p \in \mathbb{R}$ (note that we didn't use the restriction $p > 0$).

□

Exercise 2.5. *Numerical integration* (more on Week 4) refers to numerical computation of integrals. Let $x_k = k/n$. The *trapezoidal rule*

$$Q_n(f) = \frac{1}{n}\sum_{k=0}^{n-1} \frac{f(x_{k+1}) + f(x_k)}{2}$$

approximates the integral $\int_0^1 f(x)\,\mathrm{d}x$. Implement the trapezoidal rule in a programming language of your choice. Then

(a) Use the trapezoidal rule the approximate the integrals of the functions

$$f(x) = \exp(-x^2) + x^4 \quad \text{and} \quad f(x) = \exp(-|x - \tfrac{1}{2}|^{1/4})$$

with $n \in \{1, 2, \ldots, 100\}$. For each of these functions, plot the *integration error*

$$\mathrm{err}(n) = \left| Q_n(f) - \int_0^1 f(x)\,\mathrm{d}x \right|$$

as a function of n. That is, on the x-axis you should have n and on the y-axis $\mathrm{err}(n)$. Use logarithmic y-axis [`matplotlib.pyplot.yscale("log")` in Python, `semilogy` in MATLAB].

The "ground truths" $\int_0^1 f(x)\,\mathrm{d}x$ do not have nice closed-form expressions. To compute $\int_0^1 f(x)\,\mathrm{d}x$, use some library function, such as `scipy.integrate.quad` or MATLAB's `integral`:

https://docs.scipy.org/doc/scipy/tutorial/integrate.html

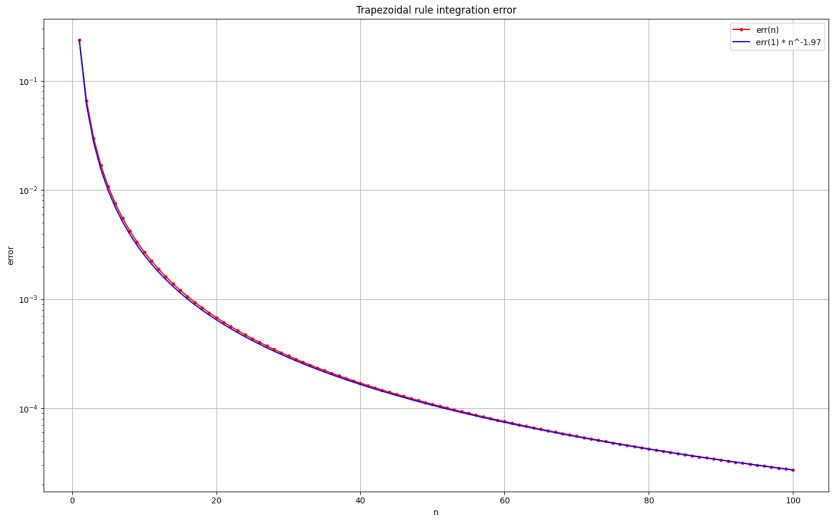
https://mathworks.com/help/matlab/ref/integral.html

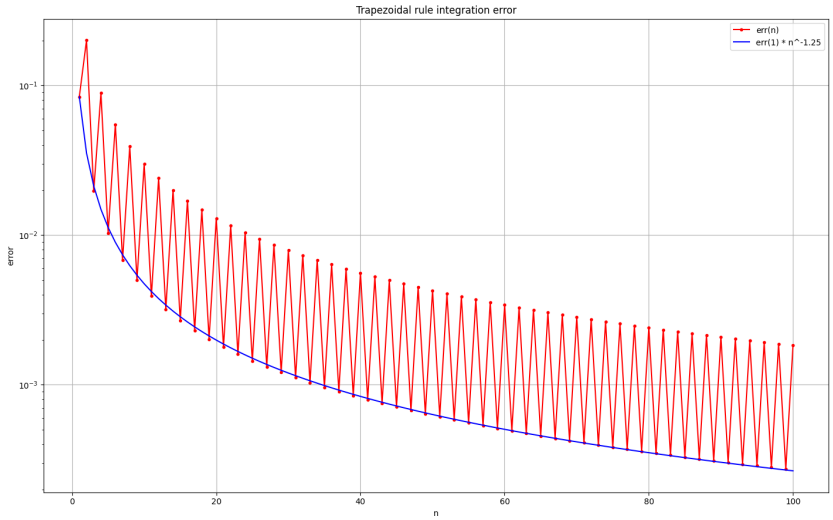
(b) Find out the *polynomial order* of the integration error for each f. That is, find constant $\alpha > 0$ such that $\mathrm{err}(n) \approx \mathrm{err}(1) \cdot n^{-\alpha}$. You can do his by trial error by plotting

err$(1) \cdot n^{-\alpha}$ for different α's and seeing which α yields a graph that approximately agrees with that of err(n).

Solution.

```
import numpy as np
import scipy.integrate as integrate
import matplotlib.pyplot as plt


def trapez(f, n, a=0, b=1):
    x = np.linspace(a, b, n + 1)
    I = np.sum((f(x[1:]) + f(x[:-1])) / 2) * (b - a) / n
    return I


# constants
N = np.arange(1, 100 + 1)


exercises = list()

f = lambda x: np.exp(-(x**2)) + x**4
alpha = 1.97

exercises.append((f, alpha))

f = lambda x: np.exp(-np.absolute(x - 1 / 2) ** (1 / 4))
alpha = 1.25

exercises.append((f, alpha))


# solve

for f, alpha in exercises:

    I = integrate.quad(f, a=0, b=1)[0] # integral "true" value

    Qnf = np.array([trapez(f, n) for n in N]) # numerical integration

    err_n = np.absolute(Qnf - I) # absolute error

    # plottings
    plt.figure()

    plt.plot(N, err_n, 'r.-', label="err(n)")
    plt.plot(N, err_n[0] * (1 / N**alpha), 'b-', label=f"err(1) * n^-{alpha}")

    # plot settings
    plt.xlabel("n")
    plt.ylabel("error")
    plt.title("Trapezoidal rule integration error")
    plt.yscale("log") # logarithmic y-axis
    plt.legend()
    plt.grid(True)
    plt.show()
```


Trapezoidal rule integration error
err(n)
err(1) * n^-1.97
error
n


Trapezoidal rule integration error
err(n)
err(1) * n^-1.25
error
n

□